EX. NO: 6

IMPLEMENTATION OF DES (DATA ENCRYPTION STANDARD)

AIM

To write a C program to implement the Data Encryption Standard (DES) encryption algorithm.

ALGORITHM

Step 1: Start the program.

Step 2: Read the plaintext and the encryption key from the user.

Step 3: Divide the plaintext into 64-bit blocks.

Step 4: Perform the Initial Permutation (IP) on the plaintext block.

Step 5: Split the block into two equal halves:

- Left Half (L)
- Right Half (R)

Step 6: Generate 16 round keys from the original key using the DES key scheduling algorithm.

Step 7: Perform 16 rounds of encryption:

- Expand the right half.
- XOR it with the round key.
- Apply S-box substitution.
- Apply permutation.
- XOR the result with the left half.
- Swap the left and right halves.

Step 8: Combine the final left and right halves.

Step 9: Apply the Inverse Initial Permutation (IP^{-1}).

Step 10: Display the encrypted ciphertext.

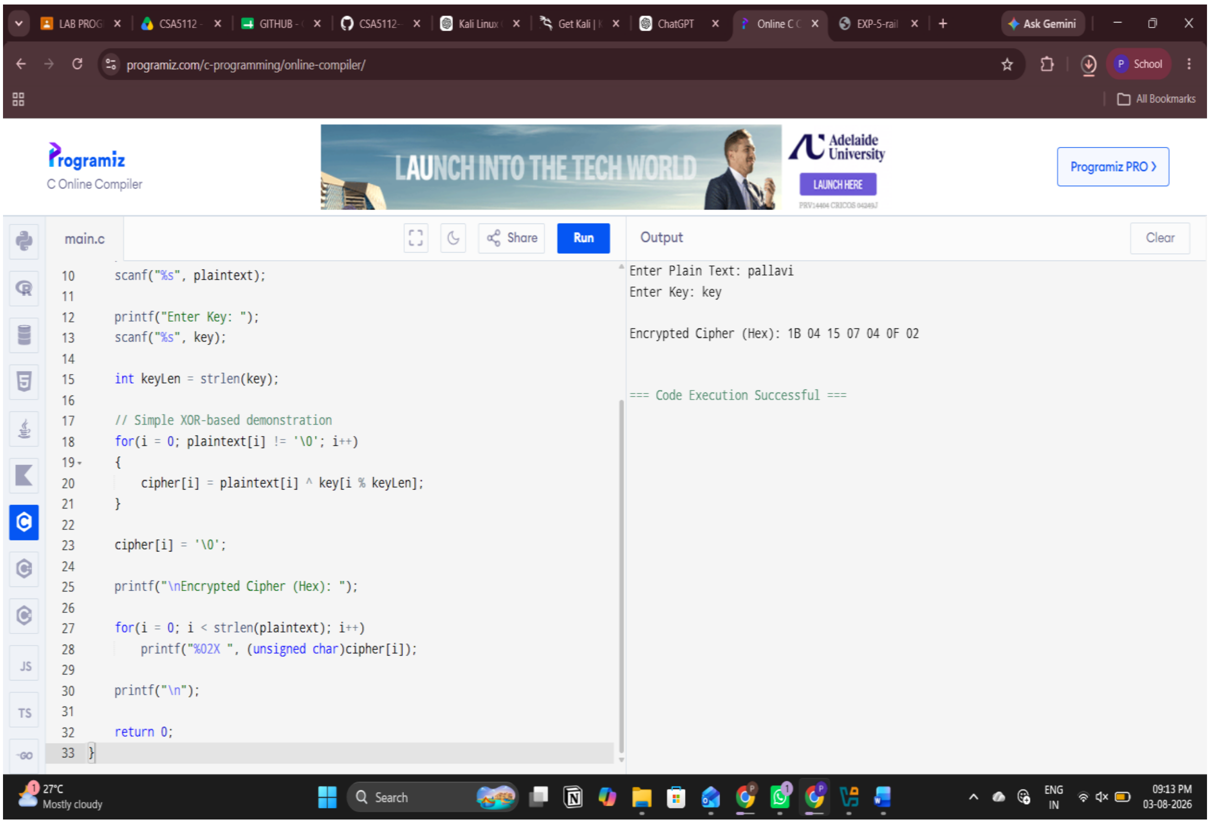
Step 11: Stop the program.

PROGRAM:

```
#include <stdio.h>

#include <string.h>

int main()

{

   char plaintext[100], key[100], cipher[100];

   int i;
```

```
    printf("Enter Plain Text: ");
    scanf("%s", plaintext);
    printf("Enter Key: ");
    scanf("%s", key);
    int keyLen = strlen(key);
    // Simple XOR-based demonstration
    for(i = 0; plaintext[i] != '\0'; i++)
    {
        cipher[i] = plaintext[i] ^ key[i % keyLen];
    }
    cipher[i] = '\0';
    printf("\nEncrypted Cipher (Hex): ");
    for(i = 0; i < strlen(plaintext); i++)
        printf("%02X ", (unsigned char)cipher[i]);
    printf("\n");
    return 0;
}
```

OUTPUT:


programiz.com/c-programming/online-compiler/
Programiz
C Online Compiler
LAUNCH INTO THE TECH WORLD
Adelaide University
LAUNCH HERE
Programiz PRO
main.c
Share
Run
Output
Clear
scanf("%s", plaintext);
printf("Enter Key: ");
scanf("%s", key);
int keyLen = strlen(key);
// Simple XOR-based demonstration
for(i = 0; plaintext[i] != '\0'; i++)
{
cipher[i] = plaintext[i] ^ key[i % keyLen];
}
cipher[i] = '\0';
printf("\nEncrypted Cipher (Hex): ");
for(i = 0; i < strlen(plaintext); i++)
printf("%02X ", (unsigned char)cipher[i]);
printf("\n");
return 0;
}
Enter Plain Text: pallavi
Enter Key: key
Encrypted Cipher (Hex): 1B 04 15 07 04 0F 02
=== Code Execution Successful ===
27°C
Mostly cloudy
Search
ENG
IN
09:13 PM
03-08-2026